21-242 Recitation 12

Fields and Vector Spaces — Q1. Unique additive identity

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State and prove that $\vec{0} \in V$ is the unique additive identity.

Course-level solution

Statement. There is a unique vector $\vec{0} \in V$ such that $\vec{v} + \vec{0} = \vec{v}$ for every $\vec{v} \in V$.
Existence is an axiom of a vector space.
Uniqueness. Suppose $\vec{e} \in V$ also satisfies $\vec{v} + \vec{e} = \vec{v}$ for every $\vec{v} \in V$. Taking $\vec{v} = \vec{0}$ and using that $\vec{0}$ is an additive identity gives

$$\vec{e} = \vec{0} + \vec{e} = \vec{0}.$$

Thus the additive identity is unique.

Later-course solution (using ideas from the rest of the handout)

The same one-line uniqueness argument works in any abelian group, so it is not special to vector spaces. After you know that $V \cong \mathbb{F}^n$ when $\dim V = n < \infty$, uniqueness of $\vec{0}$ in V is equivalent to uniqueness of the zero tuple in $\mathbb{F}^n$, but that is a strictly heavier route and hides that the claim is purely algebraic.